

Solutions to Year 2 Quantum Mechanics Unassessed Problem Sheet 1:

1. Substitute into the time-dependent Schroedinger equation:

$$\left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V\right) A e^{i(kx - \omega t)} = i\hbar \frac{\partial}{\partial t} A e^{i(kx - \omega t)}$$

$$\left(\frac{\hbar^2 k^2}{2m} + V\right) A e^{i(kx - \omega t)} = \hbar \omega A e^{i(kx - \omega t)}$$

is a solution with

$$k = \frac{\sqrt{2m(\hbar\omega - V)}}{\hbar}$$

Similarly substituting:

$$\left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V\right) A \sin(kx - \omega t) = i\hbar \frac{\partial}{\partial t} A \sin(kx - \omega t)$$

$$\left(\frac{\hbar^2 k^2}{2m} + V\right) A \sin(kx - \omega t) = -i\hbar \omega A \cos(kx - \omega t)$$

$$\therefore \tan(kx - \omega t) = \text{constant} \quad \text{or} \quad A = 0$$

and since tan is obviously **not** a constant that only leaves $A = 0$.

2. Substitute into Schrodinger's equation:

$$\left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x, t)\right) \Psi = i\hbar \frac{\partial}{\partial t} \Psi$$

$$\left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x, t)\right) (c_1 \psi_1 + c_2 \psi_2) = i\hbar \frac{\partial}{\partial t} (c_1 \psi_1 + c_2 \psi_2)$$

$$\left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x, t) - i\hbar \frac{\partial}{\partial t}\right) c_1 \psi_1 = \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x, t) - i\hbar \frac{\partial}{\partial t}\right) c_2 \psi_2$$

and since both ψ_1 and ψ_2 are solutions of the Schroedinger equation, both LHS and RHS are = 0, and so Ψ is indeed a solution.

3. If the proton has 10 MeV kinetic energy ($= p^2/2m$), then its momentum is $7.3 \times 10^{-20} \text{ kgms}^{-1}$ (if you convert to SI) and hence its wavenumber $k = p/\hbar$ is $6.9 \times 10^{14} \text{ m}^{-1}$. Likewise 10 MeV corresponds to an angular frequency $\omega = E/\hbar$ is $1.5 \times 10^{22} \text{ s}^{-1}$. So the free particle wavefunction is:

$$\Psi(x, t) = e^{i(kx - \omega t)} \text{ with the above values of the two constants.}$$

The probability of finding the particle at any point is proportional to $\Psi^* \Psi$, which with the above form is just = 1 independent of x , and so the probability of finding the particle is uniform in x .

4. Since the probability density is given by the modulus-squared of the wavefunction, it has the form:

$$P(x) = \frac{2}{L} \cos^2\left(\frac{\pi x}{L}\right) \quad -L/2 \leq x \leq L/2$$

$$P(x) = 0 \quad x < -L/2 \quad \text{and} \quad x > L/2.$$

Integrating we obtain:

$$\begin{aligned} \int_{-\infty}^{+\infty} P(x) dx &= \frac{2}{L} \int_{-L/2}^{+L/2} \cos^2\left(\frac{\pi x}{L}\right) dx \\ &= \frac{2}{L} \int_{-L/2}^{+L/2} \frac{1}{2} \left(1 + \cos\left(\frac{2\pi x}{L}\right)\right) dx \\ &= \frac{1}{L} \left[x + \frac{L}{2\pi} \sin\left(\frac{2\pi x}{L}\right) \right]_{-L/2}^{+L/2} = 1 \end{aligned}$$

So the wavefunction is normalised to unit probability of finding the particle *somewhere*, i.e. it has a sensible physical normalisation.

The probability that the particle is found in the region $-L/4 \leq x \leq L/4$ is

$$\begin{aligned} \frac{2}{L} \int_{-L/4}^{+L/4} \cos^2\left(\frac{\pi x}{L}\right) dx &= \frac{1}{L} \left[x + \frac{L}{2\pi} \sin\left(\frac{2\pi x}{L}\right) \right]_{-L/4}^{+L/4} = \frac{1}{L} \left[\frac{L}{2} + \frac{2L}{2\pi} \right]_{-L/4}^{+L/4} = \frac{1}{2} + \frac{1}{\pi} = 0.818 \\ \langle \hat{x} \rangle &= \int_{-L/2}^{+L/2} \sqrt{\frac{2}{L}} \cos\left(\frac{\pi x}{L}\right) \hat{x} \sqrt{\frac{2}{L}} \cos\left(\frac{\pi x}{L}\right) dx = \int_{-L/2}^{+L/2} x \frac{2}{L} \cos^2\left(\frac{\pi x}{L}\right) dx = 0 \end{aligned}$$

as overall odd function integrated between symmetric limits. Or explicitly evaluate

$$\begin{aligned} \int_{-L/2}^{+L/2} \frac{2}{L} \frac{1}{2} \left[x + x \cos\left(\frac{2\pi x}{L}\right) \right] dx &= 0 + \frac{1}{L} \left[x \left(\frac{L}{2\pi} \right) \sin \frac{2\pi x}{L} \right]_{-L/2}^{+L/2} - \frac{1}{L} \int_{-L/2}^{+L/2} \left(\frac{L}{2\pi} \right) \sin \frac{2\pi x}{L} dx \\ \langle \hat{p} \rangle &= \int_{-L/2}^{+L/2} \sqrt{\frac{2}{L}} \cos\left(\frac{\pi x}{L}\right) \hat{p} \sqrt{\frac{2}{L}} \cos\left(\frac{\pi x}{L}\right) dx = -i\hbar \frac{2}{L} \int_{-L/2}^{+L/2} \cos\left(\frac{\pi x}{L}\right) \left(\frac{\pi}{L}\right) \sin\left(\frac{\pi x}{L}\right) dx \\ \langle \hat{p} \rangle &= -i\hbar \frac{\pi}{L^2} \int_{-L/2}^{+L/2} \sin\left(\frac{2\pi x}{L}\right) dx = 0 \end{aligned}$$

Also

$$\langle \hat{p}^2 \rangle = -\hbar^2 \frac{2}{L} \int_{-L/2}^{+L/2} \cos\left(\frac{\pi x}{L}\right) \left(-\frac{\pi}{L}\right)^2 \cos\left(\frac{\pi x}{L}\right) dx = \left(\frac{\pi\hbar}{L}\right)^2 \int_{-L/2}^{+L/2} \frac{2}{L} \cos^2\left(\frac{\pi x}{L}\right) dx = \left(\frac{\pi\hbar}{L}\right)^2$$

Finally

$$\begin{aligned}
 \langle \hat{x}^2 \rangle &= \int_{-L/2}^{+L/2} \frac{2}{L} x^2 \cos^2 \left(\frac{\pi x}{L} \right) dx \\
 &= \int_{-L/2}^{+L/2} \frac{2}{L} \frac{1}{2} \left[x^2 + x^2 \cos \left(\frac{2\pi x}{L} \right) \right] dx \\
 &= + \frac{1}{L} \left[\frac{x^3}{3} \right]_{-L/2}^{+L/2} + \left[x^2 \left(\frac{L}{2\pi} \right) \sin \frac{2\pi x}{L} \right]_{-L/2}^{+L/2} - \frac{1}{L} \int_{-L/2}^{+L/2} \left(\frac{L}{2\pi} \right) 2x \sin \frac{2\pi x}{L} dx
 \end{aligned}$$

$$\langle \hat{x}^2 \rangle = \frac{L^2}{12} + 0 - \frac{1}{L} \left(\frac{L}{2\pi} \right)^2 \left[-2x \cos \frac{2\pi x}{L} \right]_{-L/2}^{+L/2} - \frac{1}{L} \int_{-L/2}^{+L/2} \left(\frac{L}{2\pi} \right)^2 2 \cos \frac{2\pi x}{L} dx$$

$$\langle \hat{x}^2 \rangle = \frac{L^2}{12} + \frac{1}{L} \left(\frac{L}{2\pi} \right)^2 \left[2x \cos \frac{2\pi x}{L} \right]_{-L/2}^{+L/2} - 0 = \frac{L^2}{12} + \left(\frac{L}{2\pi} \right)^2 \frac{1}{L} [-L - L] = \frac{L^2}{12} - \frac{L^2}{2\pi^2}$$

5. We normalise so that the integral over all space of the probability density should be unity.

Thus:

$$\begin{aligned}
 \int_{-\infty}^{+\infty} \psi^*(x) \psi(x) dx &= A^2 \int_{-\infty}^{+\infty} \left(\frac{x}{a} \right)^2 e^{-x^2/a^2} dx \\
 &= \frac{A^2}{a^2} \frac{\sqrt{\pi}}{2} a^3 = A^2 \frac{a\sqrt{\pi}}{2} = 1 \\
 \Rightarrow A &= \sqrt{\frac{2}{a\sqrt{\pi}}}
 \end{aligned}$$

As for the sketch, the probability density will be zero at $x = 0$, increase as x^2 for small x , then decrease as a Gaussian for larger x . Thus the sketch should look roughly like:

